



2. Strong interaction



STARTSEITE

EINFÜGEN

ZEICHNEN

ANSICHT



Beenden

Illustration

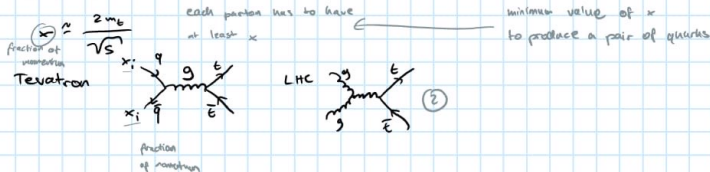
Tevatron and LHC collider

$$pp \rightarrow t\bar{t} + X \text{ LHC}$$

very low mass

$$p\bar{p} \rightarrow t\bar{t} + X \text{ Tevatron}$$

threshold production

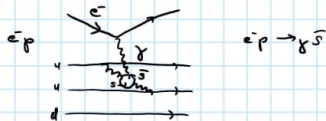
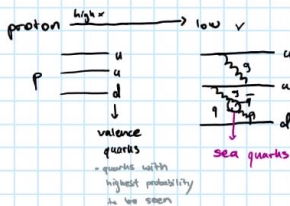


$$\text{Tevatron } \sqrt{s} = 1.96 \text{ TeV} \quad m_t \approx 175 \text{ GeV}$$

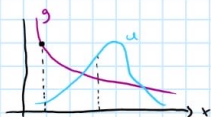
$$x \approx 0.175$$

$$\text{LHC } \sqrt{s} \approx 14 \text{ TeV}$$

$$x \approx 0.025 \Rightarrow \text{it's enough for a proton to have a 2.5\% of the momentum of the proton}$$



PDF (gluon)



$\rightarrow$ explanation why at LHC ② is more likely than ①

2) Quarks and colour

Family of quarks

$$1574 \text{ } 3/4 (c\bar{c}) \quad 895$$

$$m_{c\bar{c}} \approx 3 \text{ GeV}$$

